

ECE238L Computer Logic Design

Lab 6: State Machine for Counters

Section 003: Wednesday 2:00-4:45

GrayCode Counter VHDL Source Code, Simulation Code, and Constraints

```

8  -- Project Name:
9  -- Target Devices:
10 -- Tool Versions:
11 -- Description:
12 --
13 -- Dependencies:
14 --
15 -- Revision:
16 -- Revision 0.01 - File Created
17 -- Additional Comments:
18 --
19
20
21
22 library IEEE;
23 use IEEE.STD_LOGIC_1164.ALL;
24
25 -- Uncomment the following library declaration if using
26 -- arithmetic functions with Signed or Unsigned values
27 --use IEEE.NUMERIC_STD.ALL;
28
29 -- Uncomment the following library declaration if instantiating
30 -- any Xilinx leaf cells in this code.
31 --library UNISIM;
32 --use UNISIM.VComponents.all;
33
34 entity GrayCode is
35     Port ( Clock_System : in STD_LOGIC;
36           Reset : in STD_LOGIC;
37           Counter : out STD_LOGIC_VECTOR (1 downto 0));
38 end GrayCode;
39
40 architecture Behavioral of GrayCode is
41     component GrayCodeCounter is
42         Port ( sClock_1Hz, reset: in STD_LOGIC;
43               outputState: out STD_LOGIC_VECTOR (1 downto 0));
44     end component;
45
46     component FreqDivider is
47         Port ( Clock_System : in STD_LOGIC;
48               Clock_1Hz : out STD_LOGIC);
49     end component;
50
51     signal slowClock : STD_LOGIC := '0';
52
53 begin
54     fd : FreqDivider port map (Clock_System => Clock_System, Cloc
55     gcc : GrayCodeCounter port map (sClock_1Hz => slowClock, reset
56
57

```

```

1  set_property IOSTANDARD LVCMOS33 [get_ports {outputState[1]}]
2  set_property IOSTANDARD LVCMOS33 [get_ports {outputState[0]}]
3  set_property IOSTANDARD LVCMOS33 [get_ports {reset}]
4  set_property IOSTANDARD LVCMOS33 [get_ports {sClock_1Hz}]
5  set_property PACKAGE_PIN E3 [get_ports {sClock_1Hz}]
6  set_property PACKAGE_PIN J15 [get_ports {reset}]
7  set_property PACKAGE_PIN H17 [get_ports {outputState[0]}]
8  set_property PACKAGE_PIN K15 [get_ports {outputState[1]}]

```

```

GrayCodeCounter_tb.vhd
C:/Users/salaz25/ECE238L/GrayCodeCounter/GrayCodeCounter.srcs/sim_1/new/GrayCo

31  --library UNISIM;
32  --use UNISIM.VComponents.all;
33
34  entity GrayCodeCounter_tb is
35      -- Port ( );
36  end GrayCodeCounter_tb;
37
38  architecture Bench of GrayCodeCounter_tb is
39      component GrayCode is
40          Port ( Clock_System : in STD_LOGIC;
41                Reset : in STD_LOGIC;
42                Counter : out STD_LOGIC_VECTOR (1 downto 0));
43      end component;
44
45      signal Clock_System_tb, Reset_tb : STD_LOGIC;
46      signal Counter_tb : STD_LOGIC_VECTOR (1 downto 0);
47      constant Clock_1000MHz_Period : time := 15ns;
48
49      begin
50          uut : GrayCode port map (Clock_System => Clock_System_tb,
51                                  Reset => Reset_tb,
52                                  Counter => Counter_tb);
53
54          Clk_process : process
55          begin
56              Clock_System_tb <= '0'; wait for Clock_1000MHz_Period/4;
57              Clock_System_tb <= '1'; wait for Clock_1000MHz_Period/4;
58          end process;
59
60          stimulus : process
61          begin
62              Reset_tb <= '0'; wait for 300ns;
63              Reset_tb <= '1'; wait for 40ns;
64              Reset_tb <= '0'; wait for 600ns;
65          end process;
66      end Bench;

```

```

21  library IEEE;
22  use IEEE.STD_LOGIC_1164.ALL;
23
24
25 -- Uncomment the following library declaration if using
26 -- arithmetic functions with Signed or Unsigned values
27 --use IEEE.NUMERIC_STD.ALL;
28
29 -- Uncomment the following library declaration if instantiating
30 -- any Xilinx leaf cells in this code.
31 --library UNISIM;
32 --use UNISIM.VComponents.all;
33
34 entity GrayCodeCounter is
35     Port ( sClock_1Hz, reset: in STD_LOGIC;
36           outputState: out STD_LOGIC_VECTOR (1 downto 0));
37 end GrayCodeCounter ;
38
39 architecture Behavioral of GrayCodeCounter is
40
41     signal presentState : STD_LOGIC_VECTOR (1 downto 0) := "00";
42     signal nextState : STD_LOGIC_VECTOR (1 downto 0);
43
44     begin
45         process (sClock_1Hz, reset)
46         begin
47             if reset = '1' then
48                 presentState <= "00";
49             elsif rising_edge(sClock_1Hz) then
50                 presentState <= nextState;
51             end if;
52         end process;
53
54         outputState <= presentState;
55
56         nextState <= "01" when presentState = "00" else
57             "11" when presentState = "01" else
58             "10" when presentState = "11" else
59             "00";
60     end Behavioral;

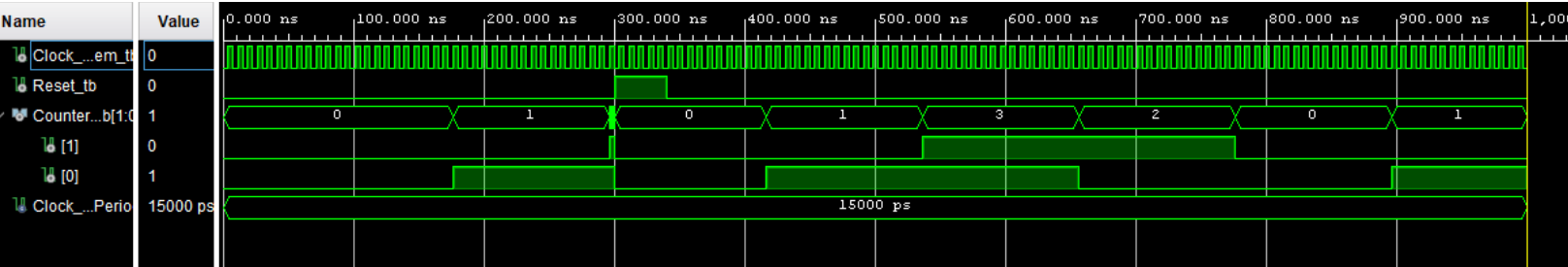
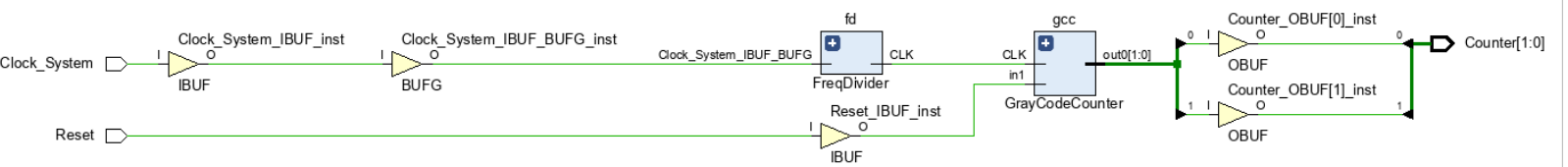
```

```

18  --
19
20
21  library IEEE;
22  use IEEE.STD_LOGIC_1164.ALL;
23  use IEEE.NUMERIC_STD.ALL;
24
25 -- Uncomment the following library declaration if using
26 -- arithmetic functions with Signed or Unsigned values
27 --use IEEE.NUMERIC_STD.ALL;
28
29 -- Uncomment the following library declaration if instantiating
30 -- any Xilinx leaf cells in this code.
31 --library UNISIM;
32 --use UNISIM.VComponents.all;
33
34 entity FreqDivider is
35     Port ( Clock_System : in STD_LOGIC;
36           Clock_1Hz : out STD_LOGIC);
37 end FreqDivider;
38
39 architecture Behavioral of FreqDivider is
40
41     signal count : INTEGER := 1 ;
42     signal sClock_1Hz : STD_LOGIC := '0';
43
44     begin
45         process (Clock_System)
46         begin
47             if Clock_System' event and Clock_System = '1' then
48                 count <= count + 1;
49                 if count = 8 then
50                     Clock_1Hz <= not sClock_1Hz;
51                     sClock_1Hz <= not sClock_1Hz;
52                     count <= 1;
53                 end if;
54             end if;
55         end process;
56     end Behavioral;
57

```

GrayCode Schematic and Wave Form



RobotEye VHDL Source Code, Simulation Code, and Constraints

The image displays three sequential screenshots of the Verilog HDL code for a project named 'RobotEye'. The code is written in a text editor with a dark background and light-colored text.

First Screenshot (Top): Shows the top of the file with comments and library declarations. The code includes comments about tool versions, descriptions, dependencies, and revisions. It also includes library declarations for IEEE, UNISIM, and UNISIM.VComponents.all.

Second Screenshot (Middle): Shows the entity and architecture declarations. The entity is named 'RobotEyeCounter_tb' and is a testbench. The architecture is named 'Bench of RobotEyeCounter_tb' and is a testbench. It includes port declarations for 'Clock_System', 'Reset', 'Cahtode7SD', and 'Anode7SD'. It also includes component declarations for 'RobotEye' and 'FreqDivider'.

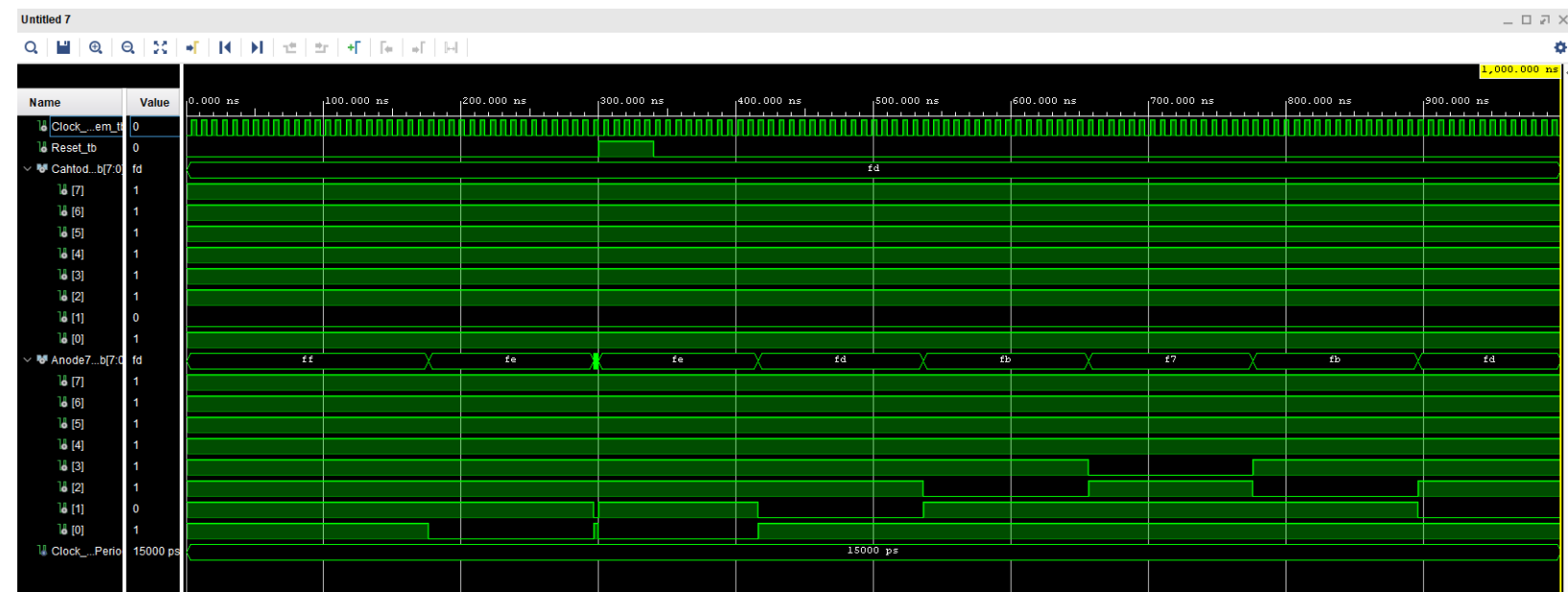
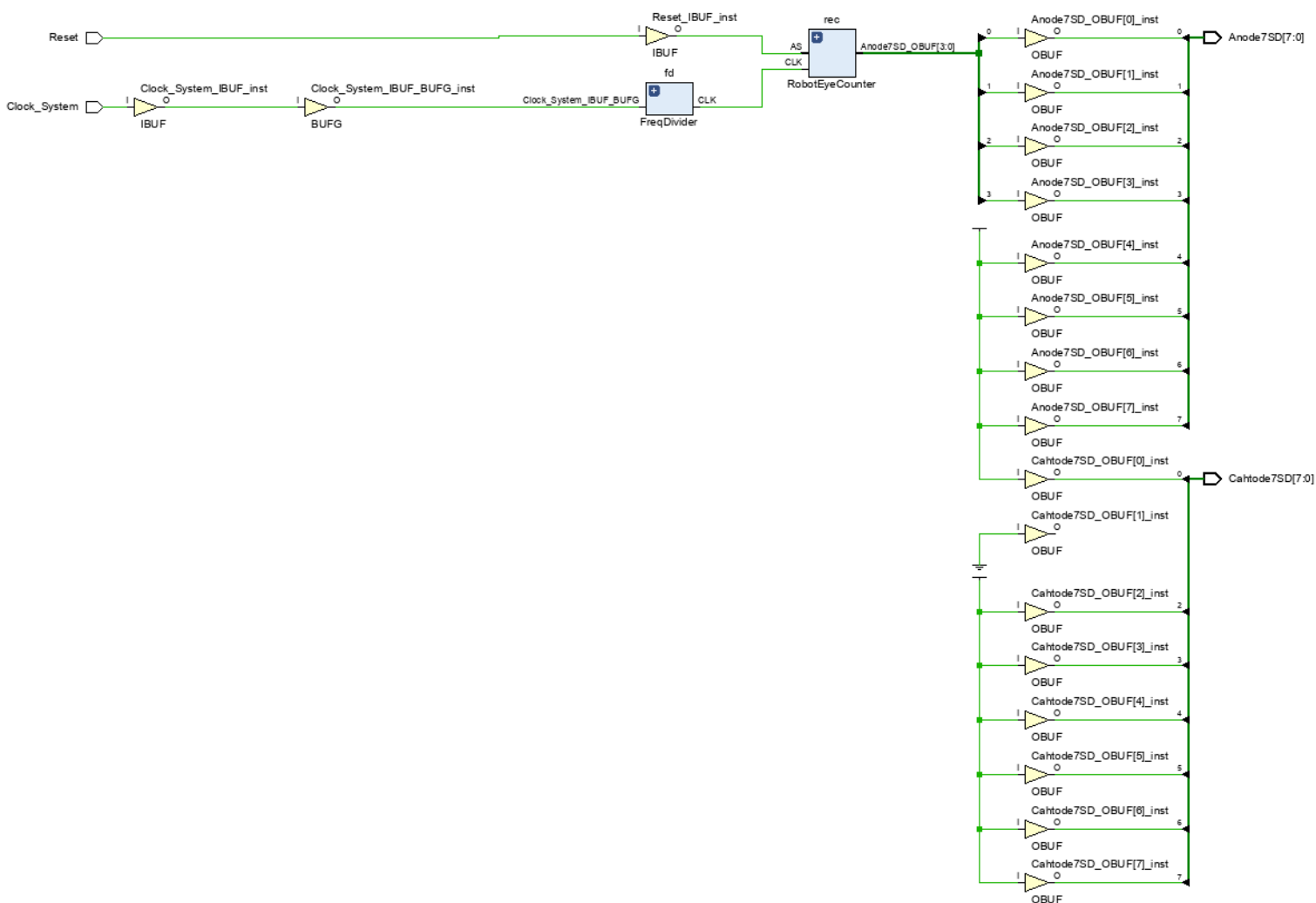
Third Screenshot (Bottom): Shows the behavioral logic for the 'FreqDivider' and 'RobotEyeCounter' components. The 'FreqDivider' component is a simple counter that outputs a square wave. The 'RobotEyeCounter' component is a more complex counter that outputs a square wave. It includes logic for the 'Clock_System' and 'Reset' signals, and for the 'Cahtode7SD' and 'Anode7SD' outputs.

```

21 library IEEE;
22 use IEEE.STD_LOGIC_1164.ALL;
23
24
25 -- Uncomment the following library declaration if using
26 -- arithmetic functions with Signed or Unsigned values
27 --use IEEE.NUMERIC_STD.ALL;
28
29 -- Uncomment the following library declaration if instantiating
30 -- any Xilinx leaf cells in this code.
31 --library UNISIM;
32 --use UNISIM.VComponents.all;
33
34 entity GrayCodeCounter is
35     Port ( sClock_1Hz, reset: in STD_LOGIC;
36           outputState: out STD_LOGIC_vector (1 downto 0));
37 end GrayCodeCounter ;
38
39 architecture Behavioral of GrayCodeCounter is
40
41     signal presentState : STD_LOGIC_vector (1 downto 0) := "00";
42     signal nextState : STD_LOGIC_vector (1 downto 0);
43
44 begin
45     process (sClock_1Hz, reset)
46     begin
47         if reset = '1' then
48             presentState <= "00";
49         elsif rising_edge(sClock_1Hz) then
50             presentState <= nextState;
51         end if;
52     end process;
53
54     outputState <= presentState;
55
56     nextState <= "01" when presentState = "00" else
57         "11" when presentState = "01" else
58         "10" when presentState = "11" else
59         "00";
60 end Behavioral;

```

RobotEye Schematic and Waveform



Timed Platform VHDL Source Code, Simulation Code, and Constraints

```
TimedPlatform.vhd
C:/Users/salaz25/ECE238L/TimePlatform/TimePlatform.srcs/sources_1/new/TimePlatform.vhd

9  -- Target Devices:
10 -- Tool Versions:
11 -- Description:
12 --
13 -- Dependencies:
14 --
15 -- Revision:
16 -- Revision 0.01 - File Created
17 -- Additional Comments:
18 --
19 -----
20
21 library IEEE;
22 use IEEE.STD_LOGIC_1164.ALL;
23
24 -- Uncomment the following library declaration if using
25 -- arithmetic functions with Signed or Unsigned values
26 --use IEEE.NUMERIC_STD.ALL;
27
28 -- Uncomment the following library declaration if instantiating
29 -- any Xilinx leaf cells in this code.
30 --library UNISIM;
31 --use UNISIM.VComponents.all;
32
33 entity TimedPlatform is
34   Port ( Clock_System : in STD_LOGIC;
35         Reset : in STD_LOGIC;
36         Cathode7SD : out STD_LOGIC_VECTOR (7 downto 0);
37         Anode7SD : out STD_LOGIC_VECTOR (7 downto 0));
38 end TimedPlatform;
39
40 architecture Behavioral of TimedPlatform is
41   component FreqDivider is
42     Port ( sClock_1Hz : in STD_LOGIC;
43           Clock_1Hz : out STD_LOGIC);
44   end component;
45
46   component TimedPlatformCounter is
47     Port ( sClock_1Hz : in STD_LOGIC;
48           Reset : in STD_LOGIC;
49           outputState : out STD_LOGIC_VECTOR (7 downto 0));
50   end component;
51
52   signal slowClock : STD_LOGIC := '0';
53
54   begin
55     fd : FreqDivider port map (Clock_System => Clock_System, Clock_1Hz => slowClock);
56     cnt : TimedPlatformCounter port map (sClock_1Hz => slowClock, Reset => Reset, outputState => Anode7SD);
57
58     Cathode7SD <= "11111101";
59
60   end Behavioral;
```

```
TimedPlatform_tb.vhd
C:/Users/salaz25/ECE238L/TimePlatform/TimePlatform.srcs/sim_1/new/TimePlatform_tb.vhd

27 --use IEEE.NUMERIC_STD.ALL;
28
29 -- Uncomment the following library declaration if instantiating
30 -- any Xilinx leaf cells in this code.
31 --library UNISIM;
32 --use UNISIM.VComponents.all;
33
34 entity TimedPlatform_tb is
35   Port ( );
36 end TimedPlatform_tb;
37
38 architecture Bench of TimedPlatform_tb is
39   component TimedPlatform is
40     Port ( Clock_System : in STD_LOGIC;
41           Reset : in STD_LOGIC;
42           Cathode7SD : out STD_LOGIC_VECTOR (7 downto 0);
43           Anode7SD : out STD_LOGIC_VECTOR (7 downto 0));
44   end component;
45
46   signal Clock_System_tb, Reset_tb : STD_LOGIC;
47   signal Cathode7SD_tb, Anode7SD_tb : STD_LOGIC_VECTOR (7 downto 0);
48   constant Clock_1000MHz_Period : time := 15ns;
49
50   begin
51     uut : timedPlatform port map (Clock_System => Clock_System_tb,
52                                  Reset => Reset_tb,
53                                  Cathode7SD => Cathode7SD_tb,
54                                  Anode7SD => Anode7SD_tb);
55
56     Clk_process : process
57     begin
58       Clock_System_tb <= '0'; wait for Clock_1000MHz_Period/4;
59       Clock_System_tb <= '1'; wait for Clock_1000MHz_Period/4;
60     end process;
61
62     stimulus : process
63     begin
64       Reset_tb <= '0'; wait for 300ns;
65       Reset_tb <= '1'; wait for 40ns;
66       Reset_tb <= '0'; wait for 600ns;
67     end process;
68
69   end Bench;
```

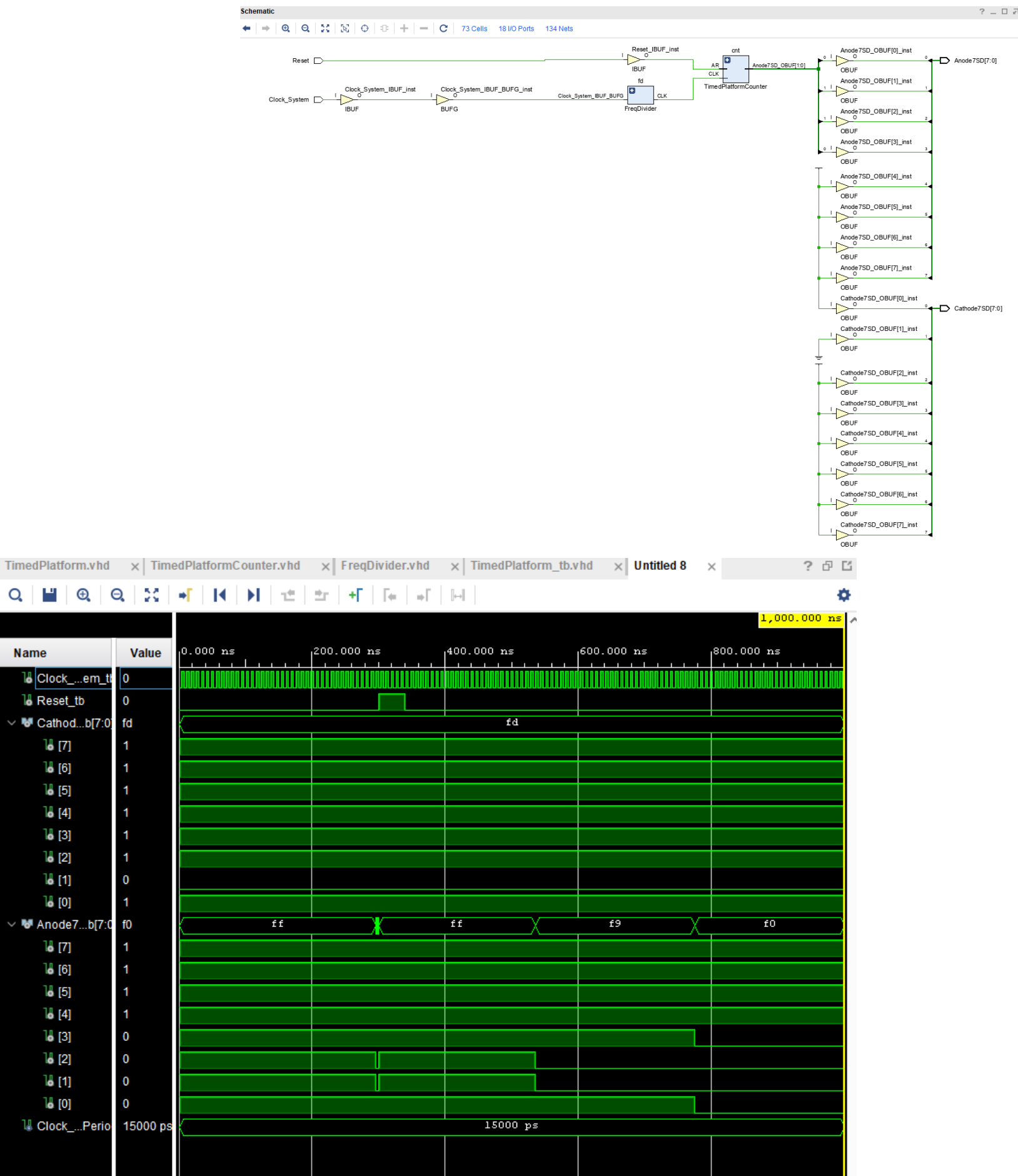
```
TimedPlatform_constr.xdc
C:/Users/salaz25/ECE238L/TimePlatform/TimePlatform.srcs/constrs_1/new/TimePlatform_constr.xdc

1  set_property IOSTANDARD LVCMOS33 [get_ports {Anode7SD[7]}]
2  set_property IOSTANDARD LVCMOS33 [get_ports {Anode7SD[6]}]
3  set_property IOSTANDARD LVCMOS33 [get_ports {Anode7SD[5]}]
4  set_property IOSTANDARD LVCMOS33 [get_ports {Anode7SD[4]}]
5  set_property IOSTANDARD LVCMOS33 [get_ports {Anode7SD[3]}]
6  set_property IOSTANDARD LVCMOS33 [get_ports {Anode7SD[2]}]
7  set_property IOSTANDARD LVCMOS33 [get_ports {Anode7SD[1]}]
8  set_property IOSTANDARD LVCMOS33 [get_ports {Anode7SD[0]}]
9  set_property IOSTANDARD LVCMOS33 [get_ports {Cathode7SD[7]}]
10 set_property IOSTANDARD LVCMOS33 [get_ports {Cathode7SD[6]}]
11 set_property IOSTANDARD LVCMOS33 [get_ports {Cathode7SD[5]}]
12 set_property IOSTANDARD LVCMOS33 [get_ports {Cathode7SD[4]}]
13 set_property IOSTANDARD LVCMOS33 [get_ports {Cathode7SD[3]}]
14 set_property IOSTANDARD LVCMOS33 [get_ports {Cathode7SD[2]}]
15 set_property IOSTANDARD LVCMOS33 [get_ports {Cathode7SD[1]}]
16 set_property IOSTANDARD LVCMOS33 [get_ports {Cathode7SD[0]}]
17 set_property IOSTANDARD LVCMOS33 [get_ports {Clock_System}]
18 set_property IOSTANDARD LVCMOS33 [get_ports {Reset}]
19 set_property PACKAGE_PIN E3 [get_ports {Clock_System}]
20 set_property PACKAGE_PIN J15 [get_ports {Reset}]
21 set_property PACKAGE_PIN H15 [get_ports {Cathode7SD[0]}]
22 set_property PACKAGE_PIN L18 [get_ports {Cathode7SD[1]}]
23 set_property PACKAGE_PIN T11 [get_ports {Cathode7SD[2]}]
24 set_property PACKAGE_PIN P15 [get_ports {Cathode7SD[3]}]
25 set_property PACKAGE_PIN K13 [get_ports {Cathode7SD[4]}]
26 set_property PACKAGE_PIN K16 [get_ports {Cathode7SD[5]}]
27 set_property PACKAGE_PIN R10 [get_ports {Cathode7SD[6]}]
28 set_property PACKAGE_PIN T10 [get_ports {Cathode7SD[7]}]
29 set_property PACKAGE_PIN J17 [get_ports {Anode7SD[0]}]
30 set_property PACKAGE_PIN J18 [get_ports {Anode7SD[1]}]
31 set_property PACKAGE_PIN T9 [get_ports {Anode7SD[2]}]
32 set_property PACKAGE_PIN J14 [get_ports {Anode7SD[3]}]
33 set_property PACKAGE_PIN P14 [get_ports {Anode7SD[4]}]
34 set_property PACKAGE_PIN T14 [get_ports {Anode7SD[5]}]
35 set_property PACKAGE_PIN K2 [get_ports {Anode7SD[6]}]
36 set_property PACKAGE_PIN U13 [get_ports {Anode7SD[7]}]
37
```

```
34 entity TimedPlatformCounter is
35   Port ( sClock_1Hz : in STD_LOGIC;
36         Reset : in STD_LOGIC;
37         outputState : out STD_LOGIC_VECTOR (7 downto 0));
38 end TimedPlatformCounter;
39
40 architecture Behavioral of TimedPlatformCounter is
41   signal presentState : STD_LOGIC_VECTOR (5 downto 0) := "000000";
42   signal nextState : STD_LOGIC_VECTOR (5 downto 0);
43
44   begin
45     process (sClock_1Hz, Reset)
46     begin
47       if Reset = '1' then
48         presentState <= "000000";
49       elsif rising_edge (sClock_1Hz) then
50         presentState <= nextState;
51       end if;
52     end process;
53
54     outputState <= "1111" & not (presentState (3 downto 0));
55
56     nextState <= "100000" when presentState = "000000" else
57               "000110" when presentState = "100000" else
58               "010110" when presentState = "000110" else
59               "001111" when presentState = "010110" else
60
61               "011111" when presentState = "001111" else
62               "101111" when presentState = "011111" else
63               "111111" when presentState = "101111" else
64               "100110" when presentState = "111111" else
65
66               "110110" when presentState = "100110" else
67               "110000" when presentState = "110110" else
68               "010000" when presentState = "110000" else
69               "000000";
70
71
72
73
74 end Behavioral;
```

```
16 -- Revision 0.01 - File Created
17 -- Additional Comments:
18 --
19 -----
20
21 library IEEE;
22 use IEEE.STD_LOGIC_1164.ALL;
23 use IEEE.NUMERIC_STD.ALL;
24
25 -- Uncomment the following library declaration if using
26 -- arithmetic functions with Signed or Unsigned values
27 --use IEEE.NUMERIC_STD.ALL;
28
29 -- Uncomment the following library declaration if instantiating
30 -- any Xilinx leaf cells in this code.
31 --library UNISIM;
32 --use UNISIM.VComponents.all;
33
34 entity FreqDivider is
35   Port ( Clock_System : in STD_LOGIC;
36         Clock_1Hz : out STD_LOGIC);
37 end FreqDivider;
38
39 architecture Behavioral of FreqDivider is
40   signal count : INTEGER := 1;
41   signal sClock_1Hz : STD_LOGIC := '0';
42
43   begin
44     process (Clock_System)
45     begin
46       if Clock_System' event and Clock_System = '1' then
47         count <= count + 1;
48         if count = 50000000 then --LINE FOR CHANGING
49           Clock_1Hz <= not sClock_1Hz;
50           sClock_1Hz <= not sClock_1Hz;
51           count <= 1;
52         end if;
53       end if;
54     end process;
55   end Behavioral;
```

Timed Platform Schematic and Waveform



Summary:

All part of this lab were built on the Frequency Divider since all part were counters. All of the parts were all very similar in implementation. The Grey Code (part a), the Robot Eye Counter (part b), and the Time Platform Counter (part c) all required the same structure. All part used the same Frequency Divider and and VHDL code to implement the different types of Sequence Counters. The biggest difference was how the counter were implemented, they Grey Code used the lights on the Artrix-7, the Robot Eye and the Timed Platform used the 7-Segment Display but had different representations. Overall a good, easy lab.

Problems:

There weren't any real problems, Once the structure was understtod for part A, parts B and C were really simple.

Hints:

Use the same structure for all parts.

Recommendations:

None, Good lab.